
TCN Multi-Output ECG Forecasting

Configuration Comparison Report

Ibrahim Hanafy

Nile University — Research Internship 2026

s-ibrahim.hanafy@zewailcity.edu.eg

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1 Objective

This report compares two Temporal Convolutional Network (TCN) configurations for multi-step ECG forecasting on the MIT-BIH Arrhythmia Database. The goal is to demonstrate that our modified configuration outperforms the original paper architecture, thereby justifying systematic hyperparameter optimization as the next research phase.

2 Experimental Setup

2.1 Dataset

- **Source:** MIT-BIH Arrhythmia Database (21 patients, excluding records 111 & 118)
- **Strategy:** Multi-Output (MIMO) — single model predicts all future steps simultaneously
- **Horizons:** $H \in \{1, 10, 15, 20\}$ steps ahead

2.2 Configuration Comparison

Table 1: Hyperparameter configurations compared.

Hyperparameter	Paper Config	Our Config
Residual Blocks	1	3 (dilations: 1, 2, 4)
Filters	256	64
Batch Size	128	256
Kernel Size	3	3
Dropout Rate	0.1	0.1
Learning Rate	10^{-3}	10^{-3}
Epochs	200	200
Early Stopping	patience 50	patience 50
Optimizer	Adam	Adam

Key difference: Our configuration trades wide filters (256) for a deeper architecture (3 residual blocks with 64 filters each), increasing the effective receptive field through stacked dilations while using $4\times$ fewer parameters per layer.

3 Results

3.1 Aggregate Performance

Table 2: Mean metrics across all 21 patients. **Bold** = better.

H	RMSE ↓		MAE ↓		R² ↑	
	Ours	Paper	Ours	Paper	Ours	Paper
1	0.0052	0.0061	0.0036	0.0037	0.9970	0.9952
10	0.0304	0.0344	0.0134	0.0148	0.8991	0.8776
15	0.0433	0.0471	0.0185	0.0200	0.8108	0.7847
20	0.0539	0.0571	0.0236	0.0253	0.7157	0.6894

Our configuration achieves lower RMSE, lower MAE, and higher R² at *every* forecast horizon.

3.2 Win Rate Analysis

Table 3: Number of patients (out of 21) where our config outperforms.

Horizon	RMSE Wins	R² Wins
1	12/21 (57%)	12/21 (57%)
10	19/21 (90%)	19/21 (90%)
15	20/21 (95%)	20/21 (95%)
20	19/21 (90%)	19/21 (90%)

At longer horizons ($H \geq 10$), our config wins on 90–95% of patients, confirming that deeper dilated architectures generalize better for multi-step forecasting.

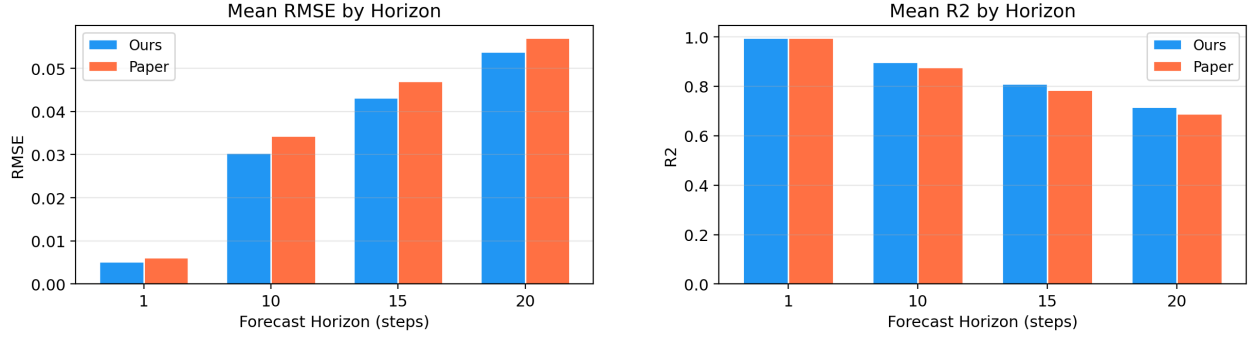
3.3 Training Efficiency

Table 4: Mean training time per patient (seconds).

Horizon	Ours (s)	Paper (s)
1	169.8	226.6
10	159.5	207.7
15	160.4	202.0
20	156.4	188.6

Our config trains 17–25% faster due to 4× fewer filters (64 vs. 256), despite having 3 residual blocks instead of 1.

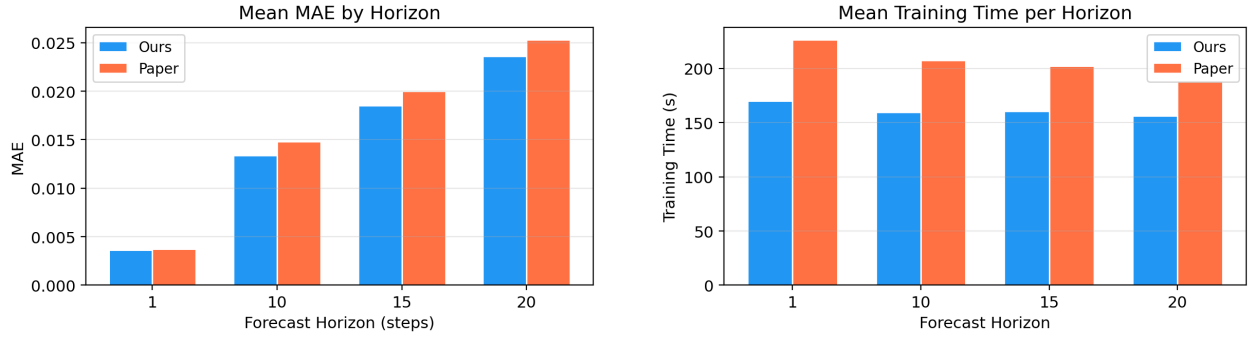
4 Visual Analysis



(a) Mean RMSE by horizon

(b) Mean R^2 by horizon

Figure 1: Our configuration (blue) consistently outperforms the paper configuration (orange).



(a) Mean MAE by horizon

(b) Mean training time by horizon

Figure 2: Our config achieves lower MAE while training faster.

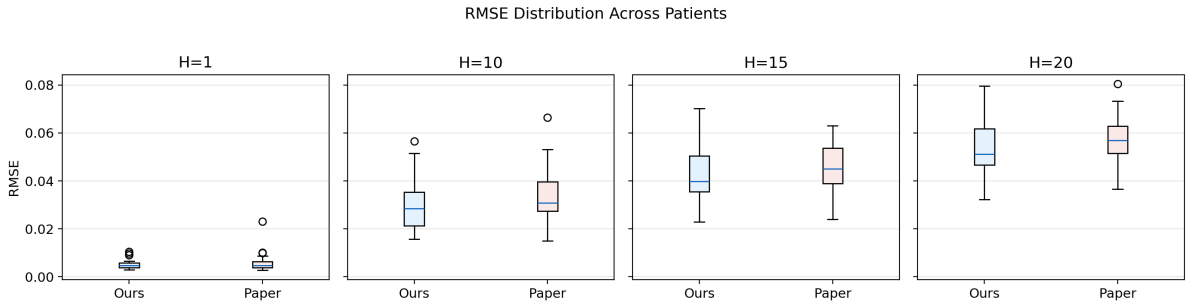


Figure 3: RMSE distribution across patients for each horizon. Our config shows tighter distributions (lower variance) and lower medians.

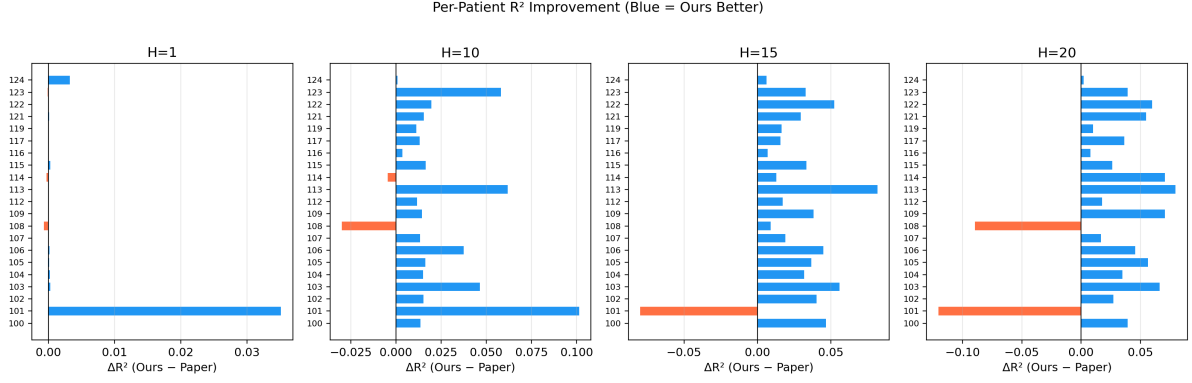


Figure 4: Per-patient $\Delta R^2 = \text{Ours} - \text{Paper}$. Blue bars indicate patients where our config achieves higher R^2 . At $H \geq 10$, nearly all patients benefit.

5 Key Findings

1. **Depth over width:** 3 residual blocks with 64 filters outperforms 1 block with 256 filters across all metrics and horizons.
2. **Stronger at longer horizons:** The advantage grows from 57% win rate at $H = 1$ to 90–95% at $H \geq 10$, indicating deeper dilated convolutions better capture long-range temporal dependencies.
3. **Computational efficiency:** Our config trains 17–25% faster due to fewer parameters, making it more practical for large-scale patient-wise experiments.
4. **Consistent improvement:** Lower variance in RMSE distributions suggests our architecture generalizes more reliably across different patient morphologies.

6 Recommendation: Hyperparameter Optimization

These results confirm that even a manual adjustment (depth vs. width) yields meaningful gains. This motivates a **systematic hyperparameter optimization** phase to explore the full configuration space:

- **Search space:** Number of blocks (2–6), filters (32–256), kernel size (2–7), dropout (0.05–0.3), learning rate (10^{-4} to 10^{-2}), batch size (64–512)
- **Methods:** Bayesian optimization (Optuna), random search, or Hyperband
- **Objective:** Minimize mean RMSE across horizons, with R^2 as secondary metric
- **Validation:** Patient-wise cross-validation to ensure generalizability